Active Directory IT Support Home Lab
Created by: Shiksha Kandel
Project Type: Active Directory and IT Support Administration Lab

1. Objective

This project involved building a Windows Active Directory lab in a virtualized environment to simulate common enterprise IT support and system administration tasks. The lab included a Windows Server configured as a Domain Controller, a Windows client machine, Organizational Units (OUs), security groups, user account management, shared folder permissions, and basic Group Policy configuration.

2. Lab Environment

- Oracle VirtualBox
- Windows Server (DC01)
- Windows 10/11 (CLIENT01)
- Active Directory Domain Services (AD DS)
- Domain: corp.local
- Host-only virtual network

3. Architecture Overview

The lab environment consists of a Windows Server configured as a Domain Controller (DC01) and a Windows client machine (CLIENT01) connected through a Host-only virtual network. The Domain Controller provides authentication, DNS resolution, and centralized directory services, while the client system acts as a domain-joined workstation used to validate user access, policy enforcement, and administrative tasks.

4. Virtual Machine and Network Configuration

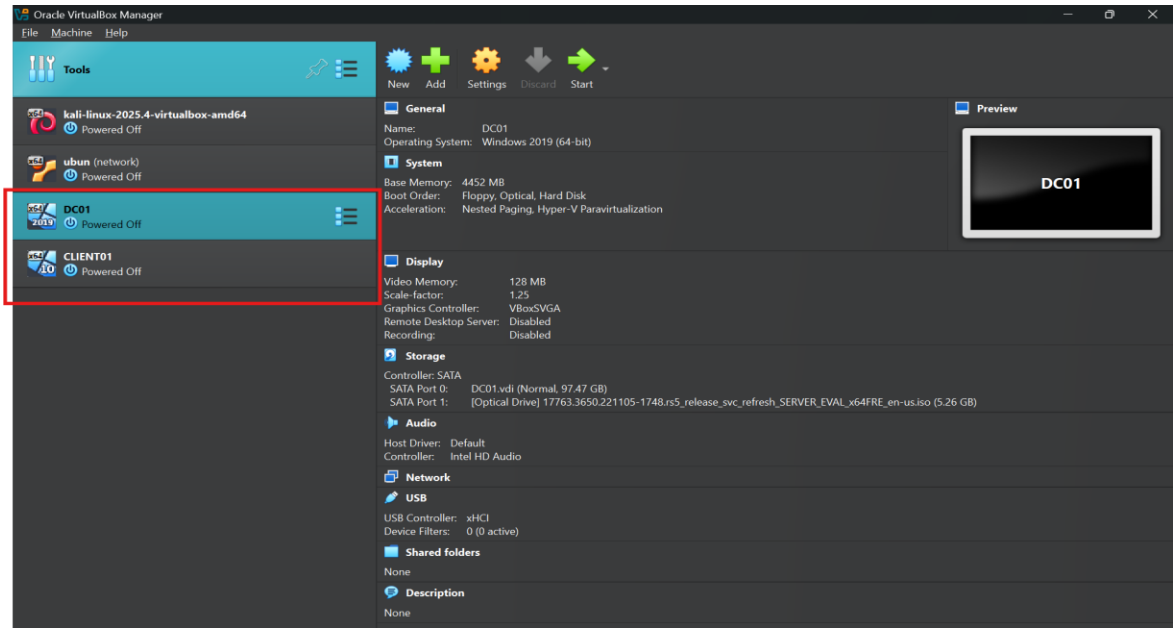


Figure 1. VirtualBox environment used to build the Active Directory lab.

This lab was built in Oracle VirtualBox using a Windows Server virtual machine for the Domain Controller and a Windows client virtual machine for endpoint administration testing.

Figure 2. Windows Server virtual machine DC01 created for the domain controller role.

A Windows Server virtual machine named DC01 was created to host Active Directory Domain Services and simulate an enterprise domain controller.

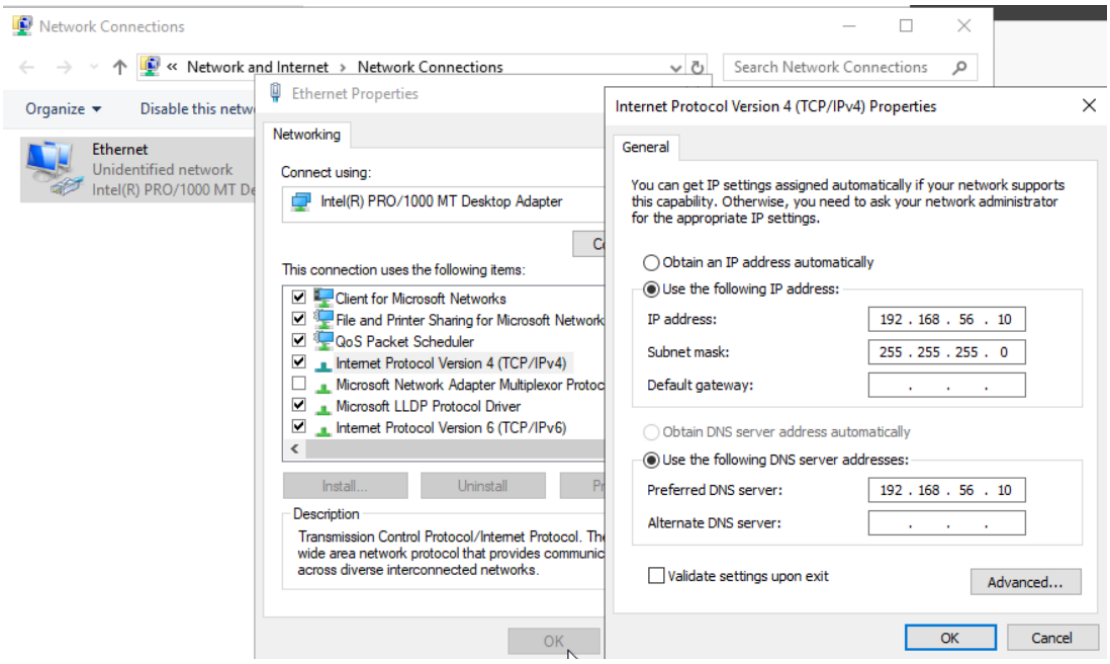


Figure 3. Static IPv4 configuration applied to DC01.

A static IP address was assigned to ensure the server could reliably function as a Domain Controller and DNS server within the lab environment.

5. Active Directory Deployment

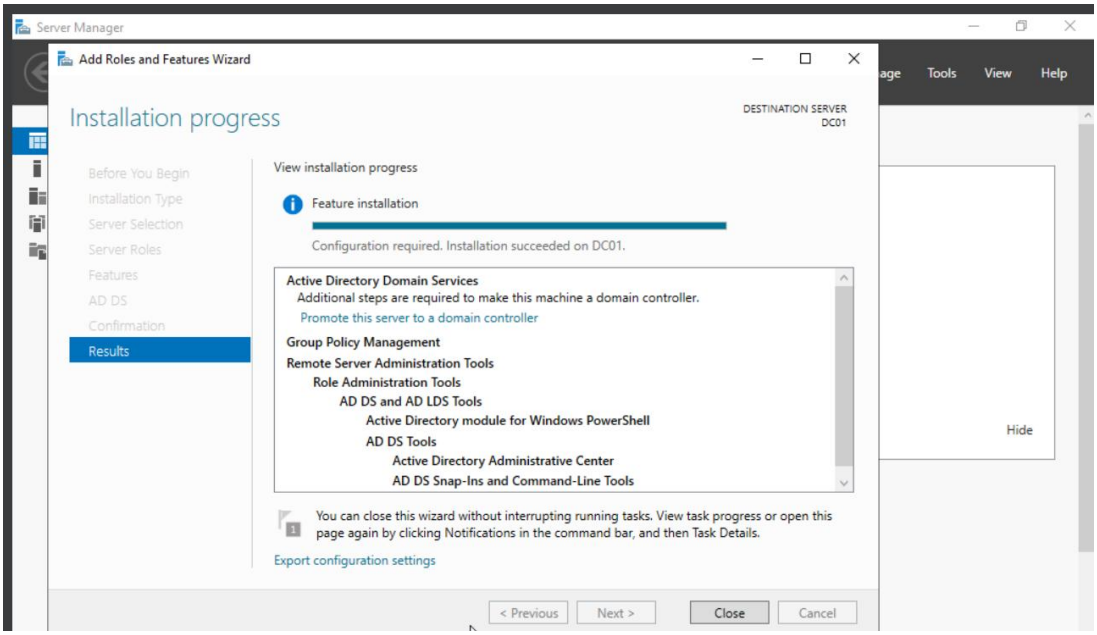


Figure 4. Active Directory Domain Services installed successfully on DC01.

Active Directory Domain Services was installed to enable centralized identity and access management. A new Active Directory forest named corp.local was created to simulate an enterprise domain environment.

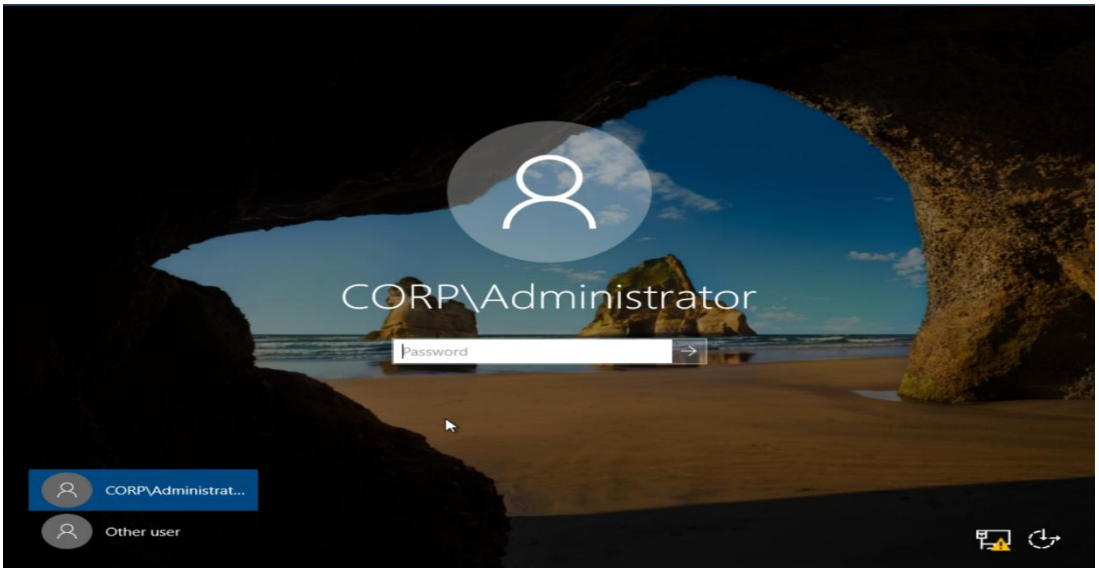


Figure 5. Successful login to the domain controller using administrative credentials.

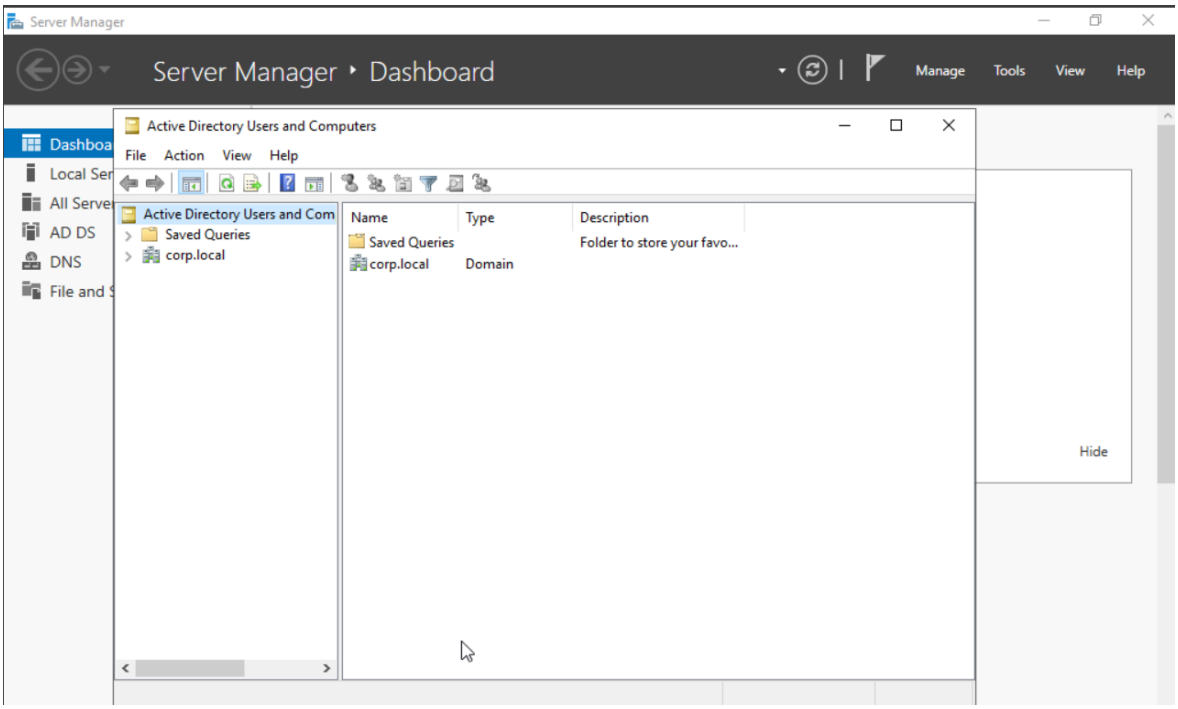


Figure 6. Active Directory Users and Computers console displaying the domain.

The ADUC console confirmed successful domain creation and readiness for administrative tasks.

6. Directory Structure and User Administration

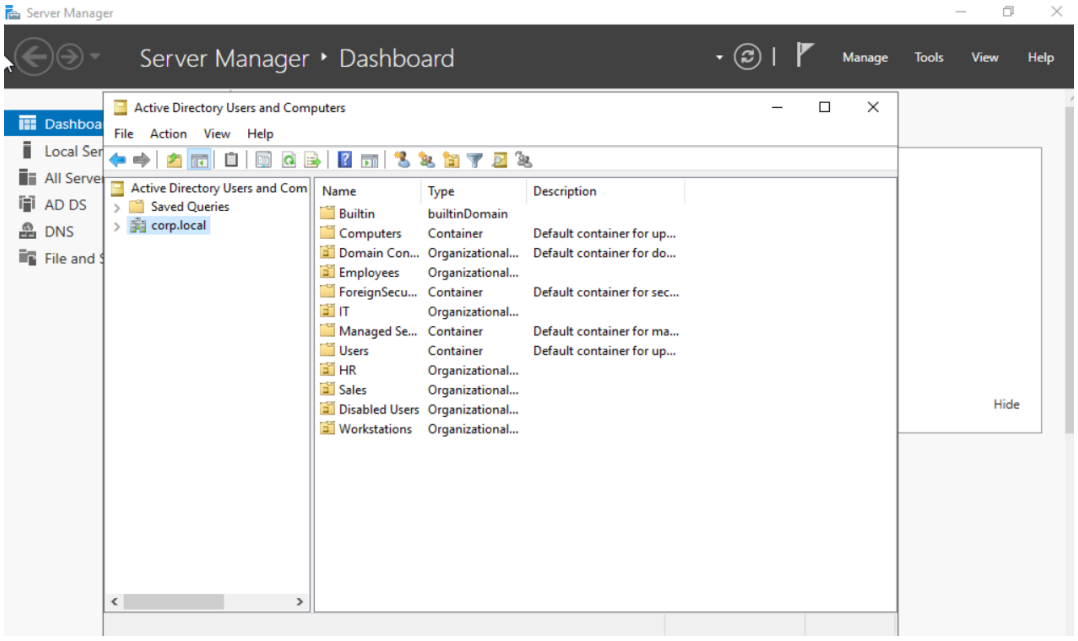


Figure 7. Organizational Units created for departmental structure.

Organizational Units were created to separate users and systems by department, improving administrative organization.

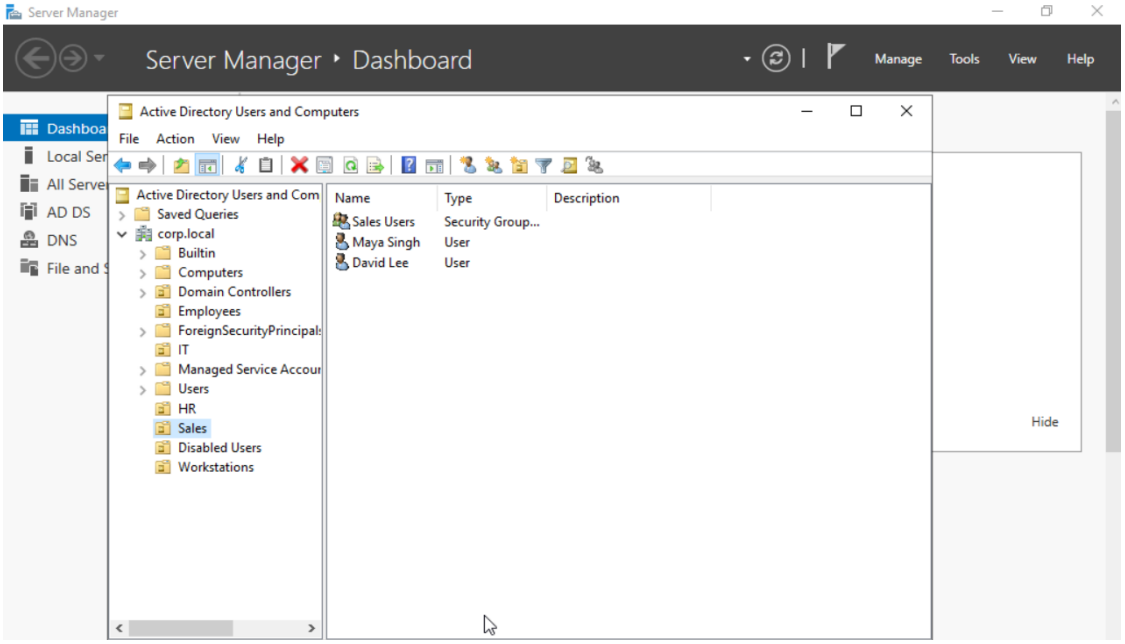


Figure 8. User accounts and security groups created within Active Directory.

User accounts and security groups were created to simulate enterprise identity management and role-based access control.

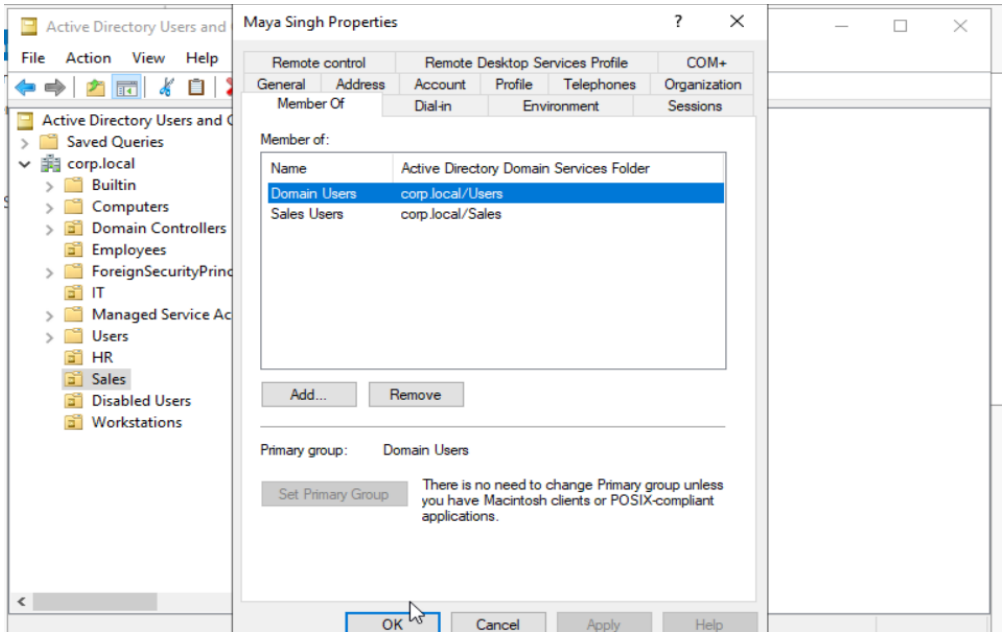


Figure 9. Group membership assigned to a domain user.

Users were assigned to department-specific groups to support controlled access and simplified permission management.

7. Client System Configuration and Domain Integration

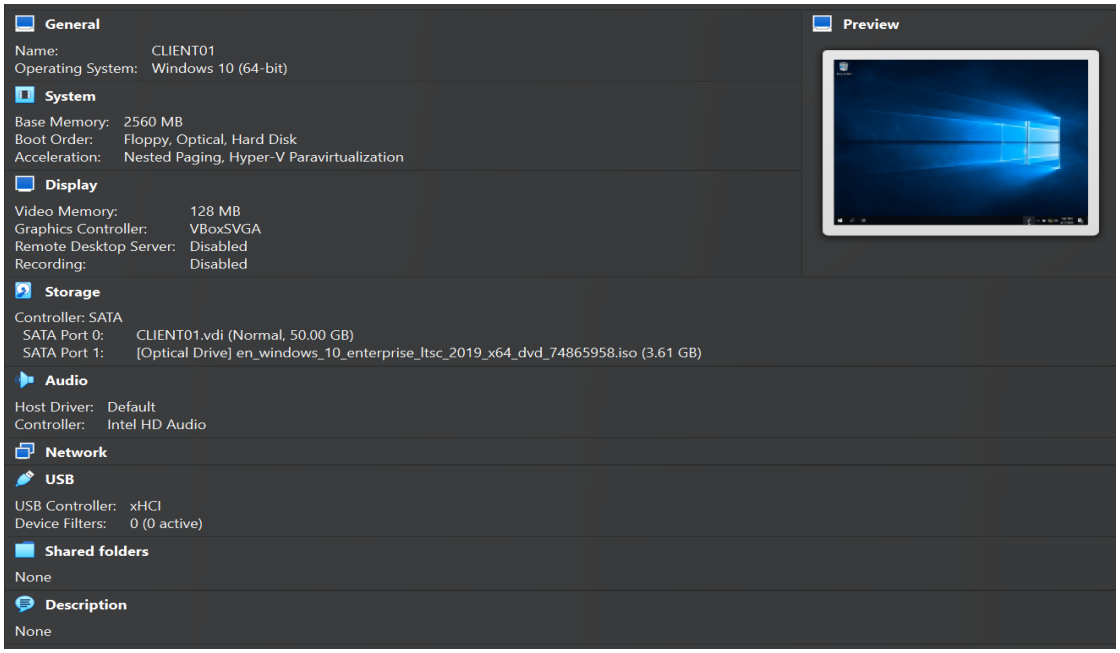


Figure 10. CLIENT01 virtual machine created to simulate a workstation.

A Windows client machine was created to represent an end-user system in the domain environment.

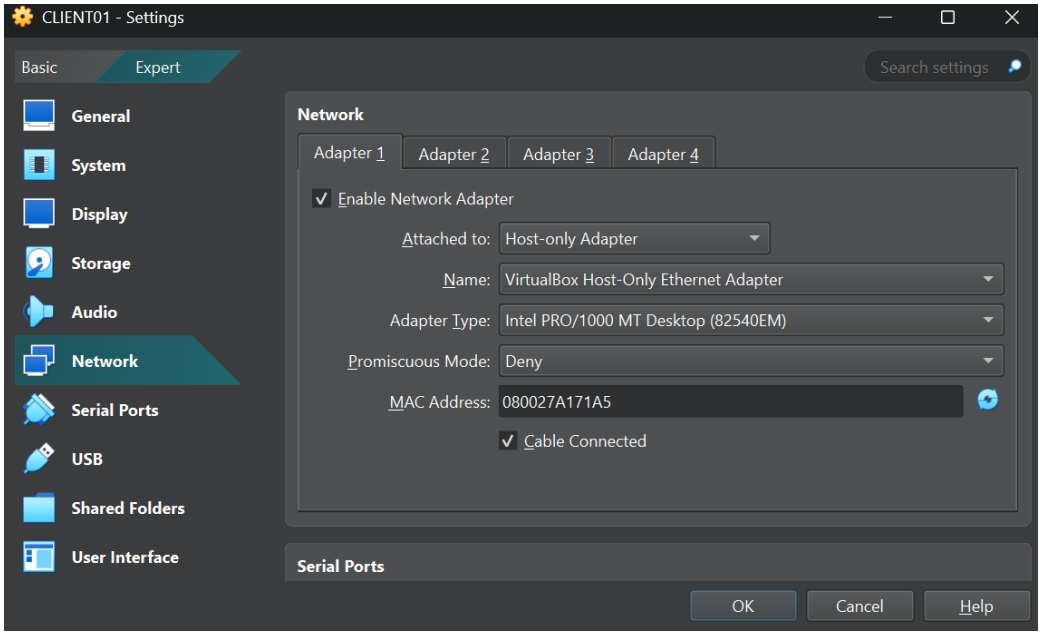


Figure 11. CLIENT01 configured on the same Host-only network as DC01.

The client was connected to the same virtual network as the Domain Controller to enable communication.

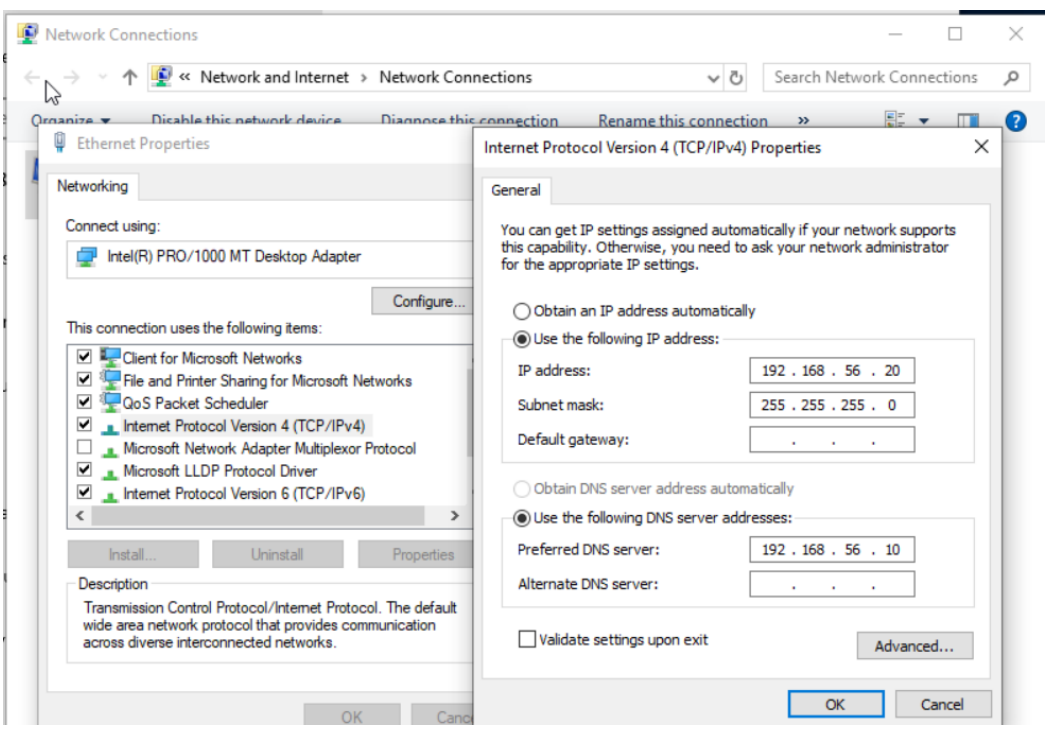


Figure 12. Static IPv4 and DNS configuration applied to CLIENT01.

The client system was configured with a static IP address and pointed to the Domain Controller for DNS resolution.

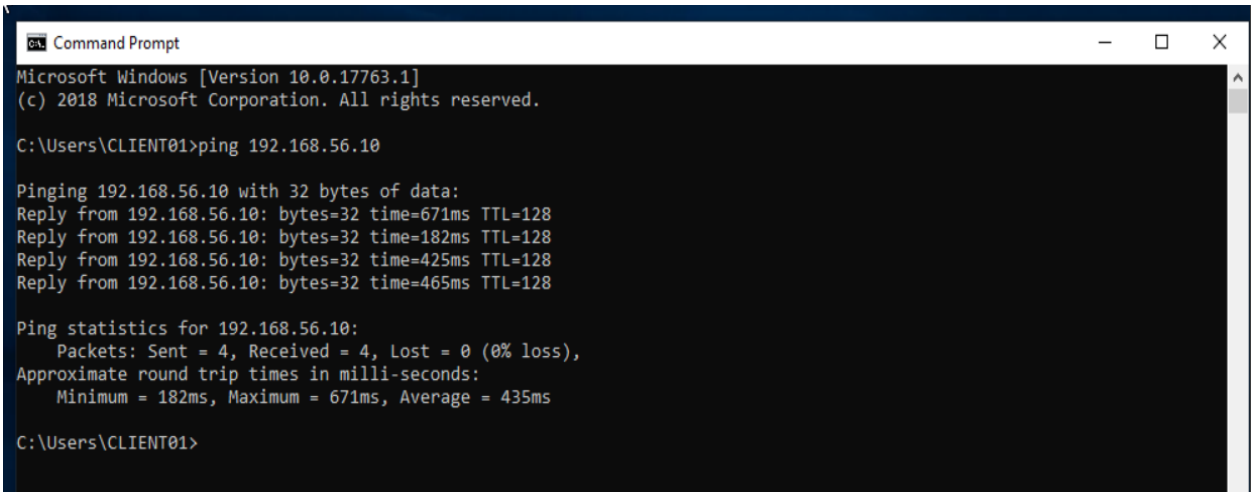


Figure 13. Successful network connectivity verified using ping.

Connectivity between CLIENT01 and DC01 was confirmed before joining the domain.

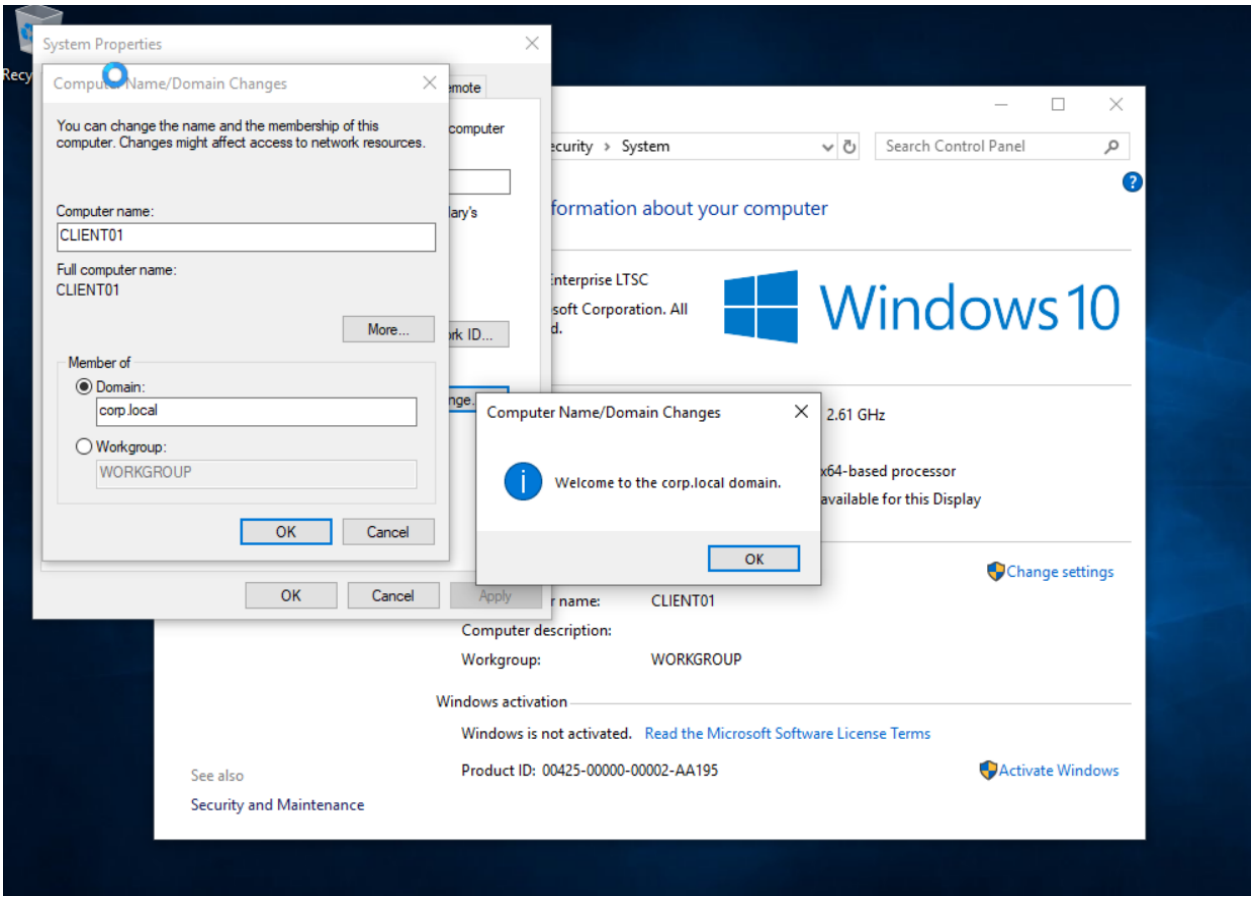


Figure 14. CLIENT01 successfully joined to the corp.local domain.

The client machine was joined to the domain using administrative credentials.

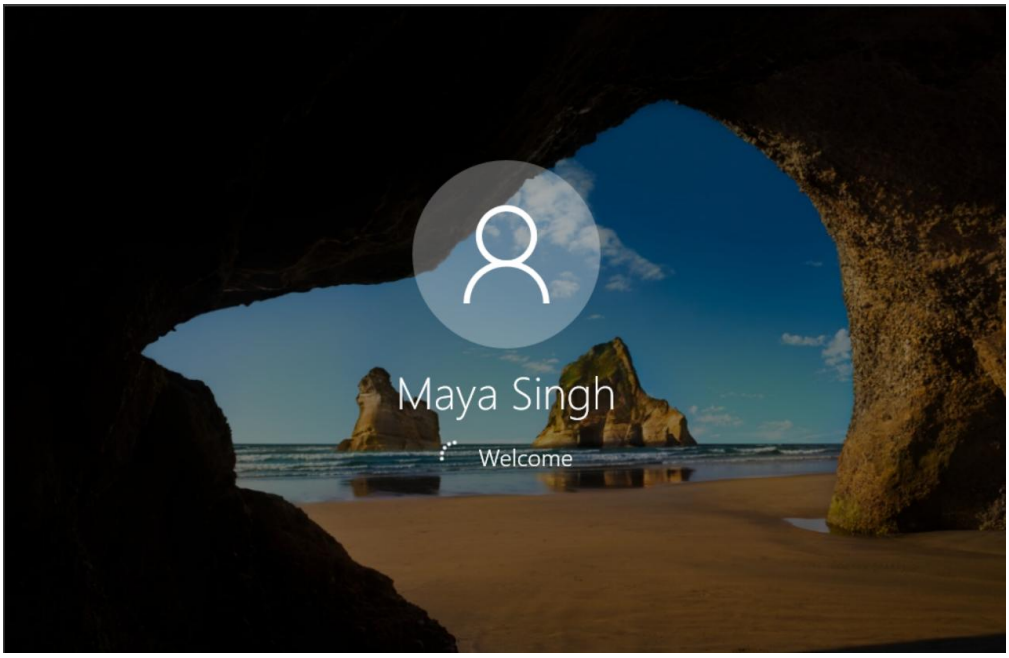


Figure 15. Successful domain user login on CLIENT01.

A domain user successfully authenticated, confirming proper domain functionality.

8. IT Support Tasks Simulation

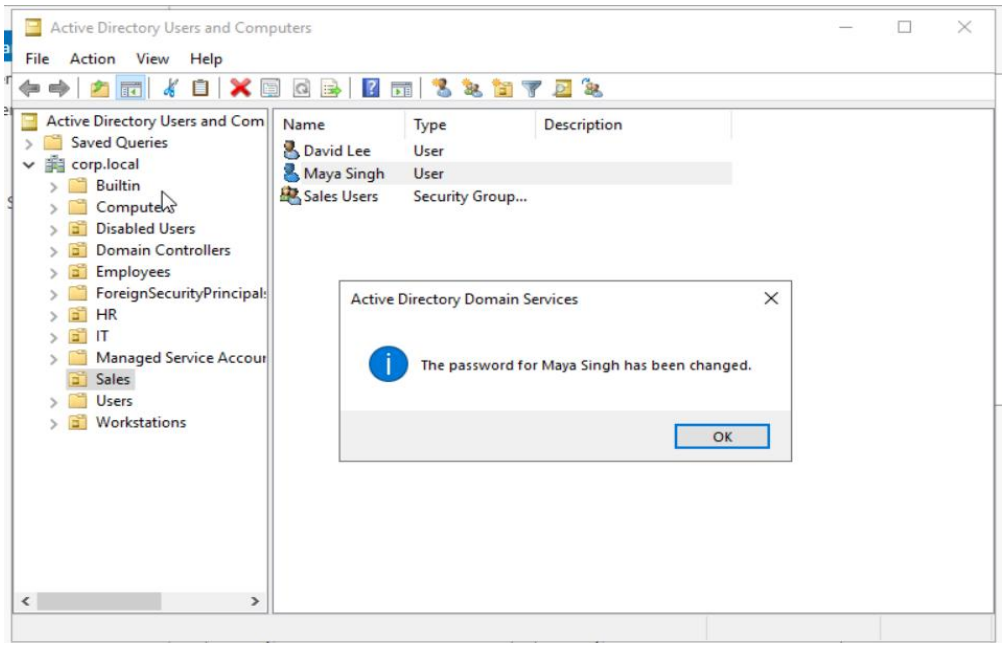


Figure 16. Password reset performed for a domain user.

A password reset was executed to simulate a common IT help desk support task.

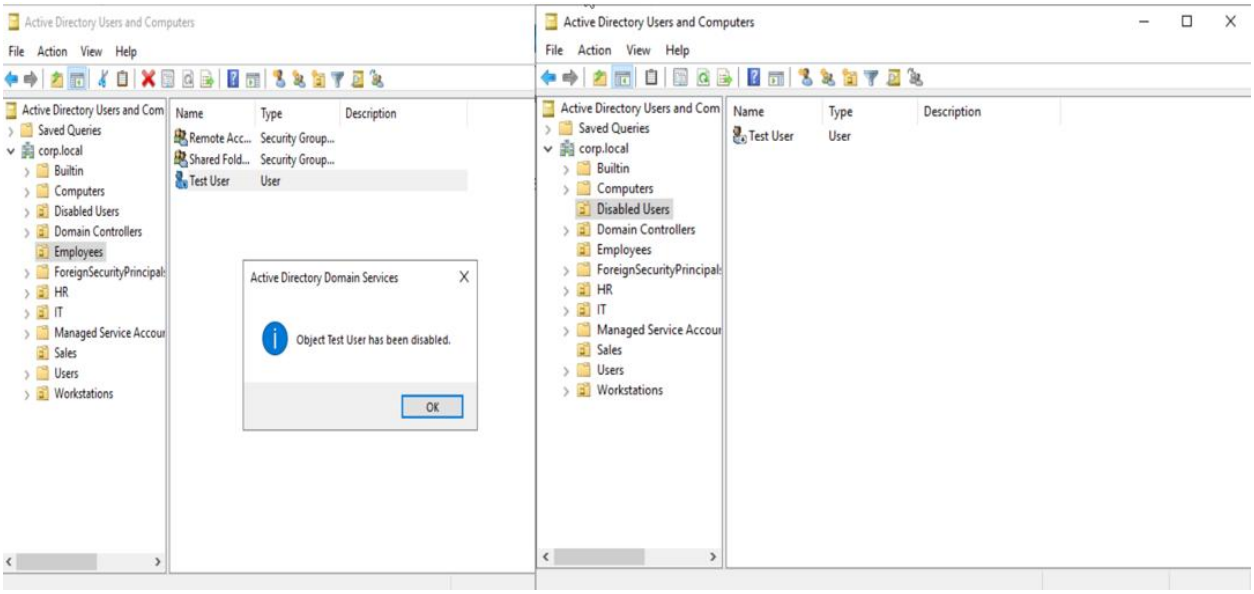
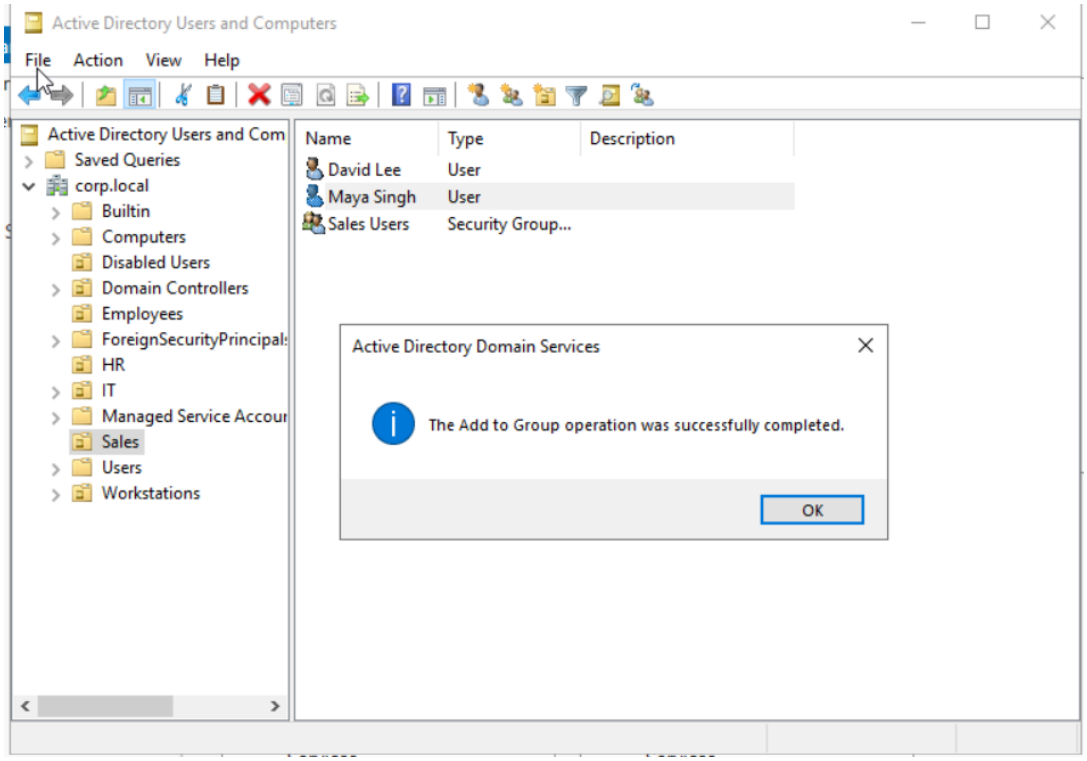


Figure 17. User account disabled and moved for offboarding.

A user account was disabled and moved to a separate OU to simulate employee offboarding procedures.

9. Shared Resource Configuration



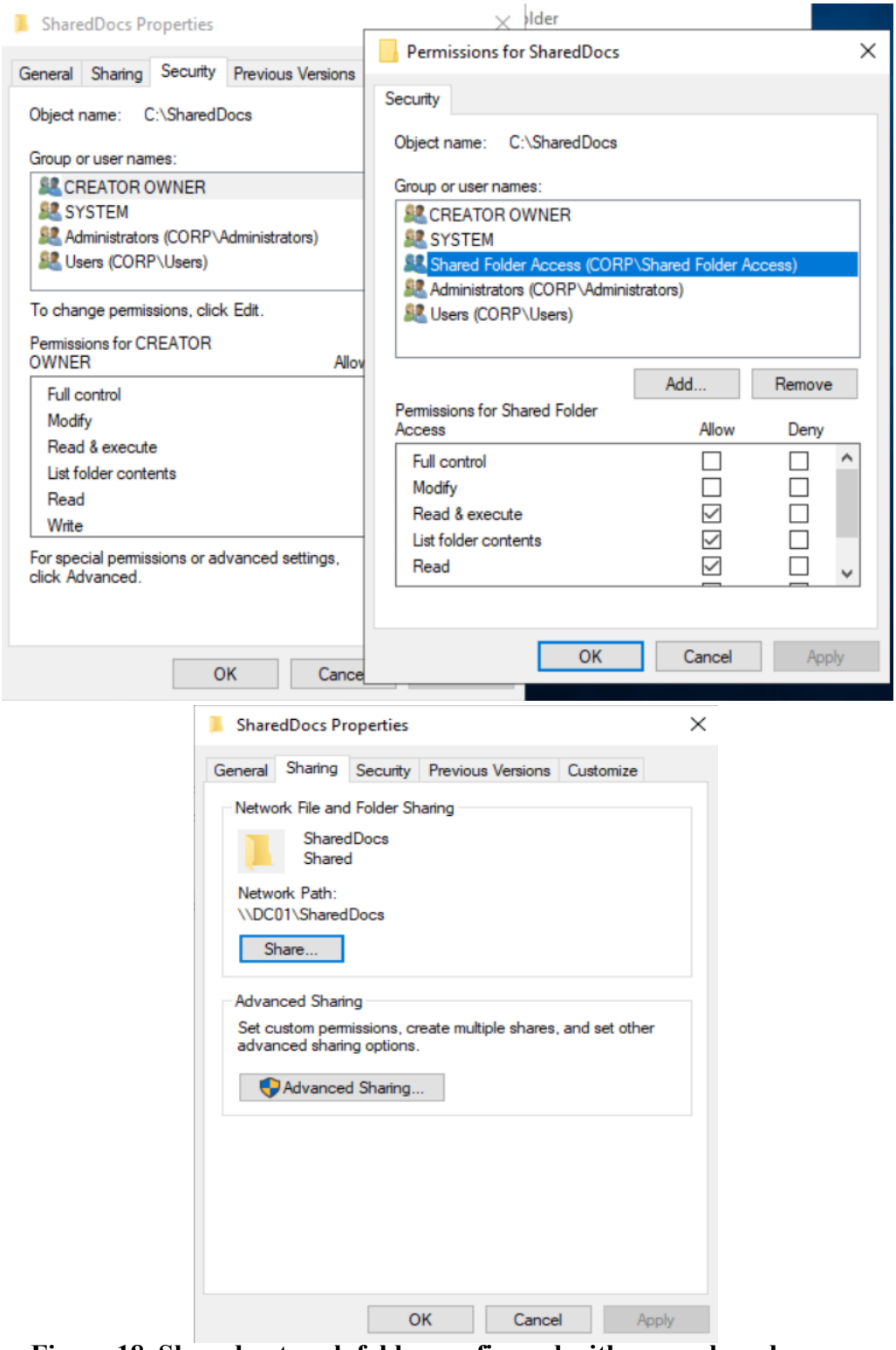


Figure 18. Shared network folder configured with group-based access.

A shared folder was created on the Domain Controller and access was granted through a security group to demonstrate centralized resource control.

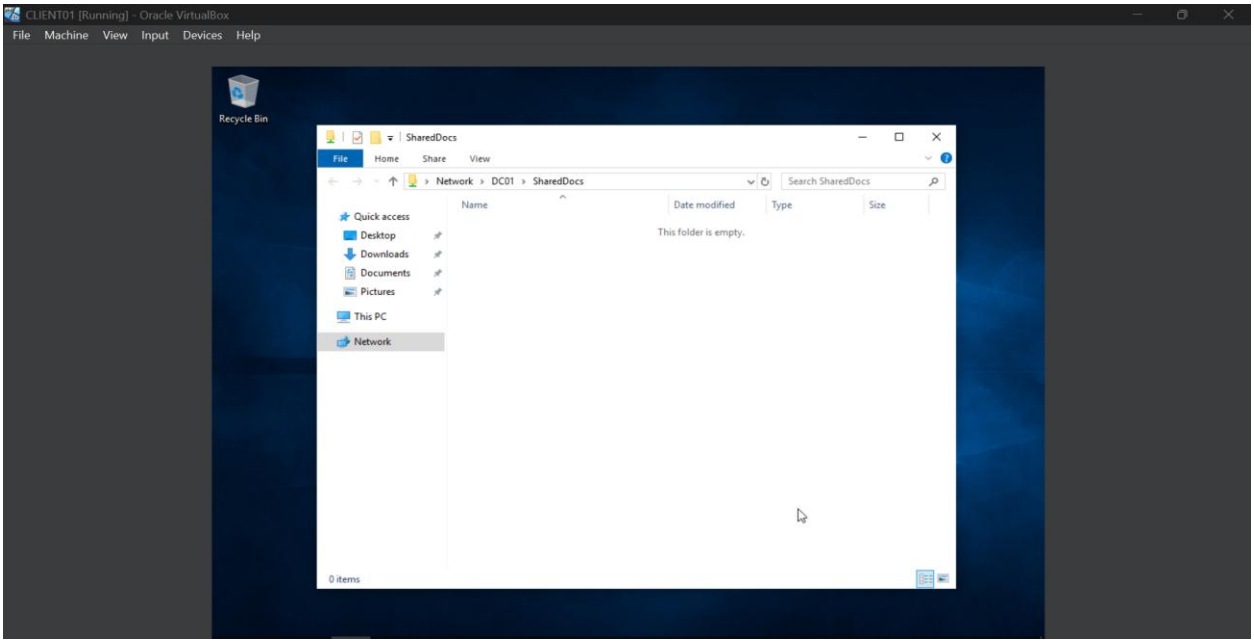


Figure 19. Successful access to the shared folder from CLIENT01.

The shared resource was accessed from the client system using a domain user account, validating permissions and connectivity.

10. Group Policy Configuration

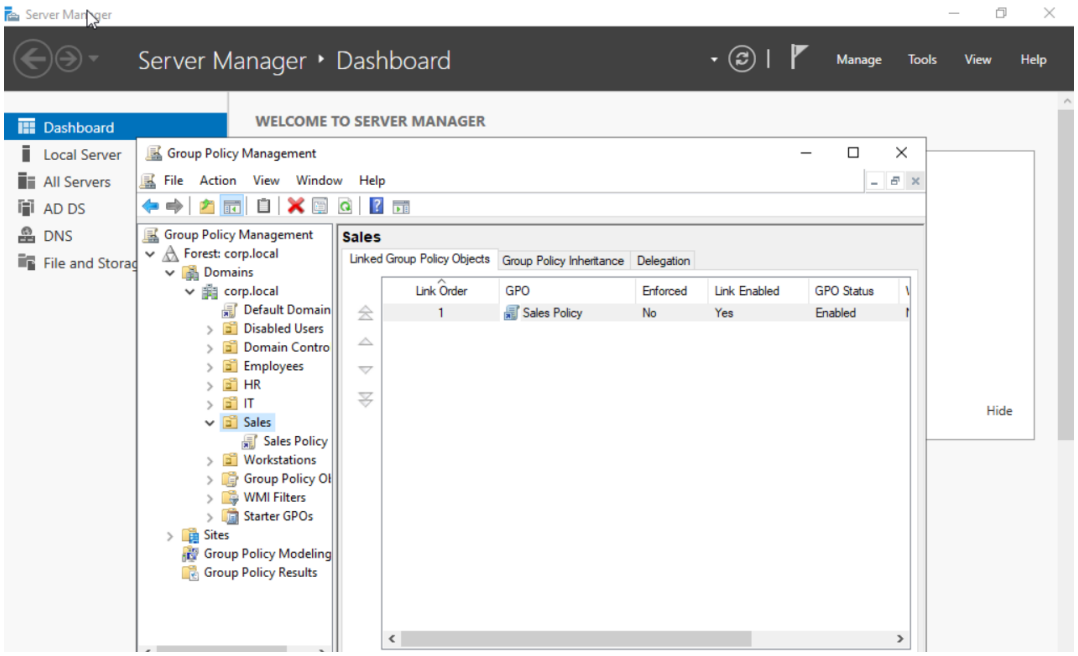


Figure 20. Basic Group Policy Object linked to an Organizational Unit.

A Group Policy Object was created and linked to a departmental OU to demonstrate centralized system management.

11. Troubleshooting and Challenges

During the setup process, issues were encountered related to domain authentication and client login delays. These were resolved by verifying DNS configuration, ensuring the client system was correctly pointed to the domain controller, and restarting the virtual machines to re-establish connectivity. This reinforced the importance of DNS in Active Directory environments.

12. Conclusion

This project demonstrated the deployment and administration of a Windows Active Directory environment in a virtual lab. A Domain Controller was configured, users and groups were created, and a client system was successfully joined to the domain. Key IT support tasks such as password resets, account management, and access control were performed. Additionally, shared resources and Group Policy were implemented to simulate centralized administration.

This lab provided practical, hands-on experience with core IT support and system administration concepts, including Active Directory management, network configuration, and troubleshooting in a domain environment.